

SUPER CODER 3.0



Name _____

Class _____ Section _____

Roll No. _____ Phone No _____

School _____



SuperCoder 3.0 | First Edition 2022 | Revised Edition 2025

As per latest guidelines given in National Skills Qualifications Framework
and the Revised Skill Education Curriculum

Aligned with NEP 2020, NCF 2023

PREFACE

In the rapidly evolving digital landscape of the 21st century, technological proficiency has become a cornerstone of modern education and professional development. As our world becomes increasingly interconnected and digitally driven, the ability to understand, navigate, and create within the digital realm is no longer a luxury, but a necessity. This shift demands a comprehensive approach to both digital literacy and coding education.

SuperCoder 3.0, developed by SuperTeacher Edureforms Pvt Ltd, offers such a curriculum designed to meet this growing need for digital competencies. This course is founded on robust educational principles, based on international standards (CSTA and ISTE), and aligned with the National Skills Qualifications Framework, the Revised Skill Education Curriculum, India's National Education Policy 2020, and the National Curriculum Framework 2023. It employs a progressive learning approach, with each level building upon the previous one.

At its core lies the **C4 model** - Context, Connection, Concept, and Competency - the load-bearing pillars that support not just the retention of information, but its practical application in the real world. This is not mere rote learning, but a living, breathing understanding of technology's place in our lives.

SuperCoder 3.0 encompasses a broad spectrum of digital knowledge areas: Digital Literacy for foundational skills, Programming and Computational Thinking for logical problem-solving, Artificial Intelligence for understanding intelligent systems, Data Science for extracting insights, Digital Citizenship and Cybersecurity for safe navigation, and Gaming and Game Development as a powerful dimension that harnesses interactive entertainment to teach design thinking, creative problem-solving, and advanced programming concepts through engaging, hands-on experiences.

It also includes skill modules recommended by experts, ensuring that the approach produces well-rounded digital citizens equipped to navigate the complexities of our society. The skills and knowledge imparted through SuperCoder 3.0 serve as the foundational elements for future innovations. They provide learners with the tools to conceptualize, design, and implement technological solutions to complex problems.

With **SuperCoder 3.0** as your blueprint, what wonders will you build? What problems will you solve? What future will you create? We invite you on this educational journey!

The SuperCoder Team
SuperTeacher Edureforms Pvt. Ltd

OUR TEAM

Authors

Panneru Bhanu Prasad
P Vishal Naidu
Vivek Avaduta
Rahul Choudhary
Subba Rao Desai

Research and Development Team

Vasudevan Natarajan
Swathi Ambati
Lukesh Sahu

Illustrations & Images

www.freepik.com

Creative Team

Mani Preetham Kuppala
Budala Romel Lymon
Chinthapally Venkatesh
Bandampally Sunny
Jindam Harshith Kumar

CONTENT FEATURES

Hi! I'm AI-Ra



I am a new creation, inspired by the A.I (Artificial Intelligence) and Ra, the Egyptian god of sun and light, both of which represent 'knowledge'. I'll be with you throughout the book, explaining the concepts in a flowing, lucid way. This textbook is written using the C4 Model: Context Setting, Connecting to Real Life, Concept Exploration, and Competency Building.



The chapters begin with context-setting: a question which is answered at the end. It can be a problem statement, or a topic-related intriguing question to spark the readers' curiosity and learning goals of the chapter in brief.

InfoByte



The InfoBytes contain interesting facts related to the topic to hook the learners to the lesson.

try < it >



There will be many 'Try it's' sprinkled in between the chapter body. They are simple activities to keep the students' minds stimulated.



This definition boxes are used whenever a new term is introduced. These define the term, without breaking the flow of the lesson.

LET'S RECAP!

'Let's Recap' summarizes the important points taught in the chapter.

WORKSHEET



The worksheet has varied types of questions, to assess the learning. The questions check the competencies of the learners, rather than their rote-learning ability.

ACTIVITY JUNCTION!

'Activity Junction' directs the students to explore and improve their skills, in context of the topics taught in the chapter.

This book series integrates NEP, SDGs, Competency-Based Education frameworks, CSTA and ISTE standards throughout its content, aligning with modern educational approaches and global standards while fostering comprehensive skill development.

DOMAIN I : DIGITAL LITERACY

1. EXCEL TO EXCEL - Advanced features of MS Excel 5

- Select, copy, & move data
- Inserting rows & columns
- Charts & graphs
- Change cell dimensions
- Formatting, autofill & formulas
- Project: Smart school elections

2. BEYOND THE REALITY - Introduction to Immersive Technology 22

- Immersive Technology
- Virtual Reality (VR)
- Metaverse
- Augmented Reality (AR)
- Mixed Reality (MR)

DOMAIN II : PROGRAMMING AND COMPUTATIONAL THINKING

3. THINK LIKE CODERS - Algorithms and Flowcharts 37

- Algorithms and algorithmic thinking
- Introduction to programming
- Project 2: Is it a leap year
- Pseudocode & Flowcharts
- Project 1: How to make a cup of coffee
- Block coding with MakeCode

4. MAKE CODE, WITH MAKECODE - Programming with Makecode 56

- Print command
- Project 2: Music Player
- Project 3: Addition of numbers
- Project 1: Hello World!
- Variables, Conditionals, and Operators
- Project 4: Dynamic background
- Loops
- Project 5: The remainder problem

5. MEET THE PYTHON - Introduction to Python 82

- Introduction to Python
- Variables
- Project 1: Simple Discount Calculator
- Getting started with Python
- Data types
- Project 2: Trip Mileage Finder
- Print
- Input statements

6. SIMPLE APP, GREAT IMPACT - Begin with App Development 105

- Apps in daily life
- Exploring the interface
- Project 2: Animal sounds
- Creating apps with MIT app inventor
- Project 1: Alphabets
- Types of apps

7. FROM A BLANK PAGE, TO A WEB PAGE! - Introduction To HTML 126

- Introduction to HTML
- Project: My Bio
- HTML tags
- Types of tags

DOMAIN III : ARTIFICIAL INTELLIGENCE

8. CAN A COMPUTER PREVENT ACCIDENTS? - Exploring Computer Vision 149

- Automation
- Domains of AI
- Artificial Intelligence (AI)
- Understanding Computer Vision (CV)

DOMAIN IV : DATA SCIENCE

9. FROM DATA TO WISDOM - Working with Data 162

- Data
- DIKW model
- Data vs Information
- Influence of data in our lives

DOMAIN V : DIGITAL CITIZENSHIP AND CYBERSECURITY

10. BEWARE OF BLACK HATS - Understanding Digital Safety 174

- Identity theft
- Phishing
- Ransomware attack
- Password theft
- Malware attack
- Preventing cyber attacks